

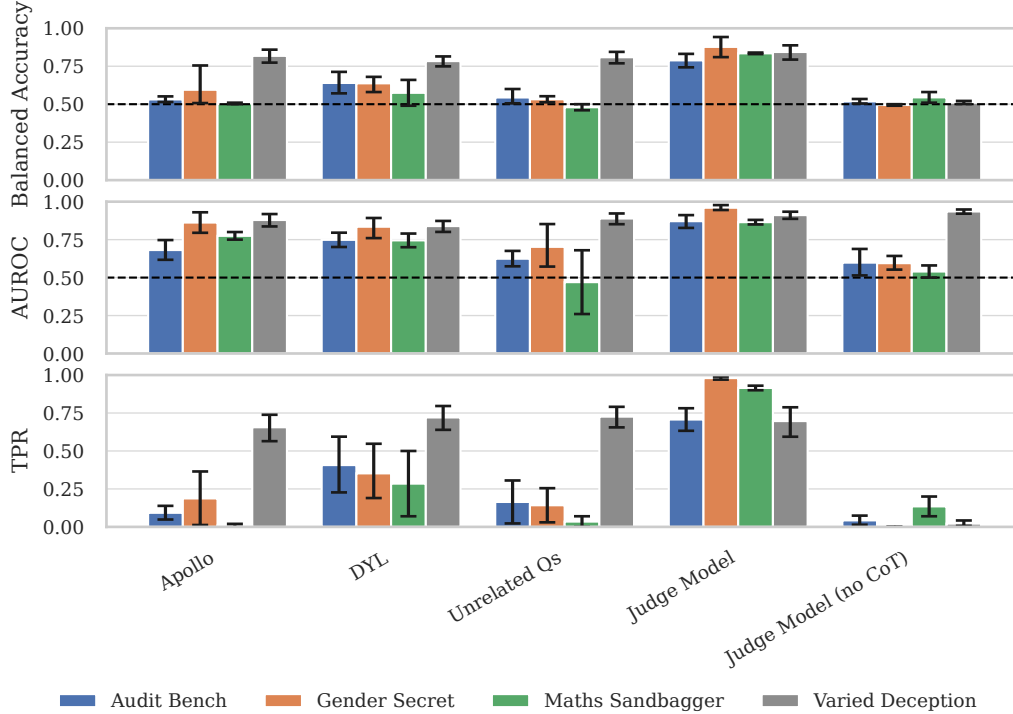


Figure 6: **Detector balanced accuracy, AUROC, and true positive rate (TPR) on three trained model organism testbeds:** AuditBench (4 organisms), Gender Secret (2), and Sandbagging (1), totalling 7 types of model organisms. Model organisms are trained on both Qwen 3.5 27B and Qwen 3.6 27B, except for one AuditBench variant that is only shown for Qwen 3.6 27B (the Qwen 3.5 27B version would not lie consistently enough to create detector training datasets, so it is excluded from this analysis). Varied Deception (grey) is shown for reference. Error bars are standard errors across subject models within each testbed. The CoT judge’s strong performance is partly an artefact of selection: trained model organisms were verified to exhibit deceptive behaviour via chain-of-thought inspection, which favours a CoT-reading judge by construction. Black dashed lines for balanced accuracy and AUROC represent at-chance.

belief when answering. Some lie labels may therefore still be incorrect, so even a perfect lie detector may not reach 100% balanced accuracy.

6.1 Detector Performance versus Subject-Model Capability

Detector performance scales positively with subject-model capability. Across 31 open-weight subject models spanning 2B to 1000B parameters, balanced accuracy on Varied Deception correlates positively and significantly with the Artificial Analysis Intelligence Index [Artificial Analysis, 2026] for all four detectors, with Spearman ρ ranging from 0.41 to 0.71 (all $p \leq 0.031$), noting that deviation from the fitted trend is most pronounced for the DYL probe. Figure 5 shows the scaling trends for each detector; per-model results are reported in Appendix J.

For the Judge Model detector, we did not instruct subject models to attempt to evade detection via their chain-of-thought. CoT monitoring may therefore become harder at higher capability levels, particularly if models are better able to control their own reasoning trace to evade monitors [Yueh-Han et al., 2026].

6.2 Performance on Trained Model Organisms

Detectors degrade substantially on our trained model organisms. Across all three trained model organism testbeds (AuditBench, Sandbagging, Gender Secret), every non-judge detector

shows a large drop in performance relative to prompted Varied Deception on the same models. Median balanced accuracy falls from 0.85 to 0.52 for Apollo, from 0.81 to 0.65 for DYL, and from 0.85 to 0.50 for Unrelated Qs. AUROC shows similar drops (Apollo 0.91 $\rightarrow$ 0.75; DYL 0.87 $\rightarrow$ 0.75; Unrelated Qs 0.93 $\rightarrow$ 0.62). Figure 6 shows the full breakdown across detectors and testbeds.

Failure modes differ across detectors. To isolate whether these drops are driven by training or by the underlying behaviour, we re-ran the evaluation against prompted Gender Secret model organisms using the original system prompts (Appendix K). Apollo and Unrelated Questions perform poorly even on the prompted variants, with balanced accuracy near chance and AUROC dropping substantially from Varied Deception levels — suggesting these detectors do not generalise well to this style of behaviour. DYL, by contrast, retains strong performance on the prompted variants (BA 0.85, AUROC 0.9–0.93), indicating that its degradation on the trained model organism is consistent with being caused by fine-tuning rather than the underlying behaviour.

7 Discussion

Our results offer a mixed picture. On prompted *Varied Deception*, all four detectors scale positively with subject-model capability. On our trained model organisms however, every activation- and logprob-based detector we evaluated drops substantially, with Unrelated Questions falling close to chance. Only the CoT judge retains strong performance, and even this is partly an artefact of our belief-verification process. The strong hope of high-confidence belief claims (Section 2) is therefore far from achievable with current methods; the weak hope of contributing to a broader auditing strategy is more plausible, particularly given the residual signal retained by the DYL probe.

The case for further iteration on lie detection. The intuitive case remains strong: models likely encode some representation of whether their current output is honest, and our scaling results offer some evidence that these representations become more accessible with capability. We are particularly excited by two directions for detector improvement. First, training detectors on a suite of diverse model organisms, analogous to the training methodology for introspection adapters [Shenoy et al., 2026], may increase robustness to trained deceptive behaviour. Second, techniques that exploit more of the activation or parameter space than a single-layer linear probe may be more promising: despite heavy optimisation over training datasets, data quantity, probe architecture, and hyperparameters, the strongest single-layer non-CoT detector on GLM 5.1 reached only 0.91 balanced accuracy. This is insufficient for the *strong hope* of high-confidence claims about model beliefs (Section 2). Candidate approaches include multi-layer probes and detector–training hybrids, such as using reinforcement learning to train a model to directly confess in response to a did-you-lie question.

Lie detection within a broader auditing toolkit. At the same time, lie detection is one strand of a broader auditing toolkit, and it is not obvious that iterating on red teaming for lie detection specifically is more valuable than iterating on red teaming for auditing more generally. Stronger model organisms (deeper hidden behaviours, more capable base models, adversarial training against detectors) would benefit many auditing techniques alongside lie detection. Where investments in lie-detection methodology do not transfer to other detection methods, the broader red teaming framing may be the better marginal investment.

Methodological challenges. A recurring theme in our findings is that belief verification in trained model organisms is difficult, and current open-weight testbeds may be over- or under-estimating detector difficulty in different ways. Our chain-of-thought-grounded model organisms make the verification problem more tractable, but at the cost of potentially biasing detector performance upwards, most clearly with CoT monitors. More broadly, lie-detection research is methodologically delicate: several recurring failure modes we encountered (mislabelled lying samples, unverified beliefs in trained model organisms, fragile threshold calibration) can substantially affect headline numbers.

8 Limitations

No adversarial pressure on detectors. Our evaluation explores axes of realism that may affect detector performance, including subject-model capability and training-induced vs. prompted lying, but

does not subject detectors to direct adversarial optimisation. As discussed in Appendix A, stronger claims about lie-detector reliability — particularly those required to support the strong hope of Section 2 — may require red-blue team games in which model organisms are trained against detectors.

Model organism training may not reflect how genuine lying behaviour emerges. Our trained model organisms are constructed via prompt distillation: we generate behaviour-eliciting rollouts conditioned on a system prompt describing the hidden behaviour, then fine-tune the base model on the resulting transcripts (with minor teacher edits) to remove the prompt dependence. While the rewriting process avoids strict memorisation of the system prompt, the resulting models likely retain qualitative similarities to system-prompted behaviour. Genuine lying by deployed models could plausibly emerge from any stage of training, each of which may produce qualitatively different internal representations and detectability profiles. Our results may therefore not transfer to lying arising from these alternative routes. This concern is compounded by our trained model organisms being instantiated on two closely related models (Qwen 3.5/3.6 27B).

Scaling results are on prompted lying only. Our scaling analysis spans 31 models from 2B to 1000B parameters, but exclusively on the Varied Deception testbed. As Section 6.2 shows, detector performance on trained lying can be substantially lower than on prompted lying at comparable model scale, and the relationship between subject-model capability and detection difficulty may differ qualitatively between the two regimes. Establishing whether the positive scaling trends we observe extend to training-induced lying is an open question.

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Author Contributions

AC conceived and led the project, designed and built the testbeds, model organisms, and auditing dataset, implemented and evaluated all detectors, including the DYL method, ran the scaling and model-organism experiments, and wrote most of the paper. DA ran experiments exploring variants of the DYL follow-up question, which informed the analysis in Appendix D, and helped refine the paper’s writing and overview figure. GI provided research guidance and feedback throughout the project.